Hubbard architecture comparison: parallel compilation and the cost of accuracy

Working research note: third refinement

22 September 2026

The shared four-step plaquette circuit does not meet the local-observable accuracy target on either of the two square patches for which exact dynamics has now been calculated. At $\tau = 1$, its doublon error is 0.01364 on 2×2 and 0.01439 on 3×3. This changes how we interpret the resource comparison: the fixed-r curves describe the cost of specified circuits, while a comparison of accurate dynamics calculations must also determine r and any spatial truncation. We do not extrapolate these small patch errors to $L = 4$ or to the bulk.

This pass also implements a parallel fermionic routing network, measures postselection bias on the eight-qubit noisy circuit, adds cultivation to both FT and hybrid state supply, and compiles explicit STAR retry control flow. These calculations identify mechanisms and remaining dependencies; they do not establish an architecture-independent NISQ limit or a settled STAR crossover.

1 Task, precision, and the comparison being made

We retain the half-filled Néel quench under

$$H/J = -\sum_{\langle ij\rangle,\sigma} (c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.}) + 4\sum_i (n_{i\uparrow} - \tfrac{1}{2})(n_{i\downarrow} - \tfrac{1}{2}), \qquad \tau = Jt, \tag{1}$$

and measure one local doublon $D = \langle n_{i\uparrow} n_{i\downarrow}\rangle$. The reference precision is 1% relative error at the provisional nonzero signal $D_* = 0.1$, hence $\epsilon = 10^{-3}$. The separate loose sensitivity is $\epsilon = 10^{-2}$ absolute. The physical array has nearest-neighbor connectivity, 10 ns gates, 100 ns readout, and $p_{2q} = 10^{-3}$. Native arbitrary-angle R_{ZZ} and one-qubit pulses of that duration, $p_{1q} = p_{2q}/10$, and idle error p per 100 ns remain declared modeling choices. Encoded reference timing retains the 240 ns syndrome cycle; it is not a calibrated physical syndrome schedule.

For a spatial region Ω, a product formula with r steps, and a noisy estimator, the useful decomposition is

$$|\widehat{D} - D_{\text{target}}| \le \epsilon_{\text{stat}} + |D_{\Omega,r} - D_\Omega| + |D_\Omega - D_{\text{target}}| + \epsilon_{\text{logical}} + \epsilon_{\text{synth}} + \epsilon_{\text{magic}}. \tag{2}$$

We split the existing 0.25ϵ product-formula reserve equally between Trotter and spatial error for the truncation scenario. Thus each receives 1.25×10^{-4} at the strict reference precision. This split replaces, rather than adds to, the old reserve. Without spatial truncation its share can be reassigned. The statistical allocation remains $\epsilon/2$ at 95% confidence, giving

$$N_0 = \left\lceil \frac{\log(2/0.05)}{2(\epsilon/2)^2} \right\rceil = 7{,}377{,}759 \tag{3}$$

ideal bounded-observable shots at strict precision.

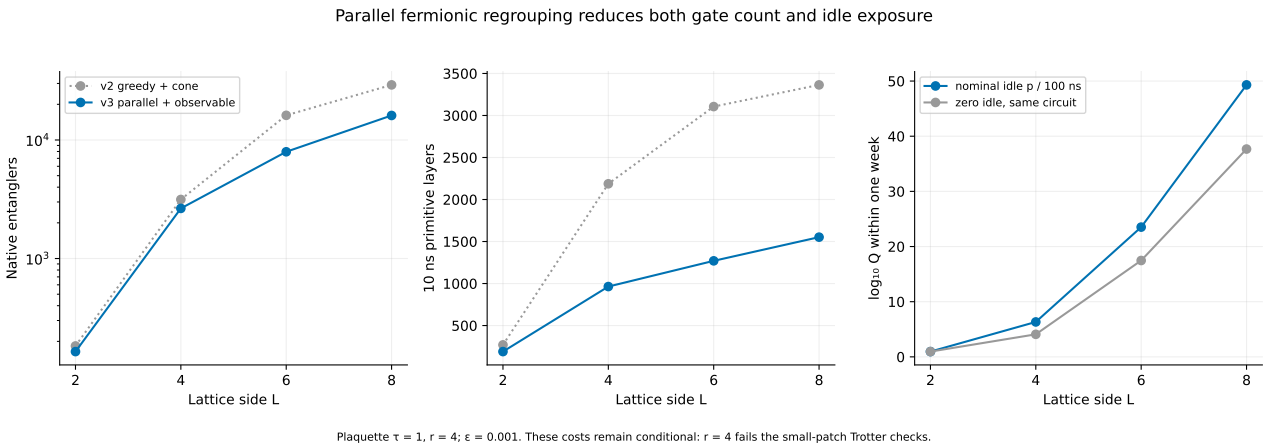


Figure 1: Compiled native counts, project-side ASAP layer counts, and the conditional week-cutoff area of the local-PEC estimator. Zero idle is a sensitivity for the same circuit, not a validated proxy for a different schedule. All panels keep $r = 4$; the accuracy checks below do not validate it.

2 Parallel regrouping on the nearest-neighbor array

The Campbell-style plaquette product formula [1] keeps its original noncommuting stage order. In each pink or gold stage, disjoint two- or four-mode fermionic groups can be gathered into contiguous intervals of the live Jordan–Wigner order. We use odd-even transposition sorting to perform this regrouping. Every transposition is an adjacent physical fermionic swap on a snake embedded in the square array. Disjoint intervals then execute their local blocks concurrently. Fermionic swap networks motivate this construction [2]; the present implementation is a heuristic and uses only the snake edges of the array.

As in the second note, number-conserving Gaussian gates on the same adjacent pair fuse into a single odd-sector $U(2)$ matrix, requiring at most two native R_{ZZ} gates plus one-qubit pulses. The compiler tries six combinations of initial mode order and local routing. A mode permutation at preparation costs no gates for the specified occupation product state. Its fermionic signs are still part of the unitary comparison, and the final measured-wire map is saved.

Two further exact identities remove boundary onsite stages. The initial occupation state is an eigenstate of H_U, so $e^{-iaH_U}|\text{Néel}\rangle$ differs only by a scalar phase. The final onsite stage commutes with the measured doublon. After removing these stages, a backwards traversal through even fermionic blocks removes blocks outside the observable's exact circuit cone. These identities do not approximate a Hamiltonian light cone. They are available to all architectures; the current encoded curves still use the earlier full logical workload.

At $L = 4$, $\tau = 1$, and $r = 4$, the selected circuit uses 2648 entanglers in 964 primitive layers, versus 3149 in 2187 layers for the v2 route. Reset and final measurement give an attempt time of 9.84 μs. The parallel full-evolution circuit, before the observable reductions, has 3204 entanglers in 1074 layers. Thus the improvement combines parallel motion, placement, fusion, and observable algebra; it is not solely a scheduling speedup. Nine selected circuits have been compiled at logical P0 in CUDA-Q, with counts cross-checked against the saved native gates.

For depolarizing noise at a k-qubit primitive, define $s_k = 1 - 4^k p_k/(4^k - 1)$. The inverse-channel PEC construction [3] has variance exposure

$$\chi = \log \Gamma^2 = \sum_{\ell \in \text{cone}} 2\log(1 + 2p_\ell/s_{k_\ell}), \qquad p_{\text{idle}}(\Delta t) = \frac{3}{4}\left[1 - (1 - 4p/3)^{\Delta t/(100\,\text{ns})}\right], \tag{4}$$

with the preparation/readout bit-flip inverses included separately. The new $L = 4$ exposure is

$$\chi = 10.5927_{2q} + 4.1201_{1q} + 5.1681_{\text{idle}} + 0.1361_{\text{SPAM}} = 20.0171. \tag{5}$$

For m independent replicas the reference time is $T \simeq N_0 e^{\chi} t_{\text{attempt}}/m$. The exact accounting uses integer replicas and the one-week cutoff. It gives $Q_{\min} = 2.13 \times 10^6$ at nominal idle noise and 12168 with zero idle. Idle exposure has decreased substantially but remains material. This is a cost of a specified sufficient estimator, not a lower bound on all possible mitigation schemes.

3 Exact patch dynamics and the accuracy–noise tradeoff

The calibration constructs H directly in fixed $(N_\uparrow, N_\downarrow)$ sectors. Sparse matrix exponentiation evaluates $e^{-i\tau H/J}$ without using the compiled gate decomposition. A separate evolution applies

$$U_r(\tau) = \left[e^{-i\Delta H_U/2} e^{-i\Delta H_{\text{pink}}/2} e^{-i\Delta H_{\text{gold}}} e^{-i\Delta H_{\text{pink}}/2} e^{-i\Delta H_U/2}\right]^r, \qquad \Delta = \tau/r, \tag{6}$$

where energies are in units of J. The largest patch is 3×3, with sector dimension $\binom{9}{4}\binom{9}{5} = 15876$. An independent eight-qubit dense Hamiltonian verifies the exact-sector observable; the original compiled plaquette circuit independently verifies the product formula on 2×2. The saved evidence contains 112 Trotter comparisons and 164 exact-curve points over four patches.

Patch, $\tau = 1$	Exact D	Error $r = 4$	Error $r = 16$	Error $r = 64$
2×2	0.1179790	0.0136397	0.00087075	0.00005447
3×3	0.1050849	0.0143939	0.00085350	0.00005317

For these square patches, $r = 64$ is the first tested power of two passing the strict split budget; $r = 16$ passes the loose split budget. These are not minimal-step claims. The asymptotic local error is consistent with r^{-2}, while coarse steps can be nonmonotonic. We also record a budget based on 1% of each actual patch signal, rather than silently replacing the provisional D_* in every resource curve.

The compiled $L = 2$ circuits at $r = 4, 16, 64$ contain respectively $164, 740, 3044$ native entanglers. The selected PEC estimator needs only one nine-qubit square copy at $r = 16$, but approximately 4.73×10^6 qubits at $r = 64$ to meet the week cutoff. Thus a common Trotter refinement can affect architecture costs very differently: it increases a physical noise exposure in an exponential, even when its algorithmic error reduction is shared. The corresponding $L = 4$ compilations demonstrate cost sensitivity only; their Trotter error has not been calibrated.

To start testing physical truncation, compare a five-site cross with its enclosing 3×3 square, keeping the same Néel pattern and central observable. Their exact differences are 4.75×10^{-4} at $\tau = 0.1$ and 7.93×10^{-2} at $\tau = 1$. Both exceed the strict truncation allowance. However,

$$|D_{\text{cross}} - D_{\text{target}}| \le |D_{\text{cross}} - D_{3\times3}| + |D_{3\times3} - D_{\text{target}}|, \tag{7}$$

and the second term is unknown. These finite-reference differences neither certify a bulk light cone nor bound the error of the larger patch. Larger patches, controlled local bounds, or both are still needed.

4 Postselection: measuring bias as well as acceptance

We propagate the full eight-qubit density matrix through the saved $L = 2$ native circuits. After each k-qubit primitive, the channel is

$$\mathcal{D}_{k,p}(\rho) = (1-p)\rho + \frac{p}{4^k - 1} \sum_{P \neq I} P\rho P. \tag{8}$$

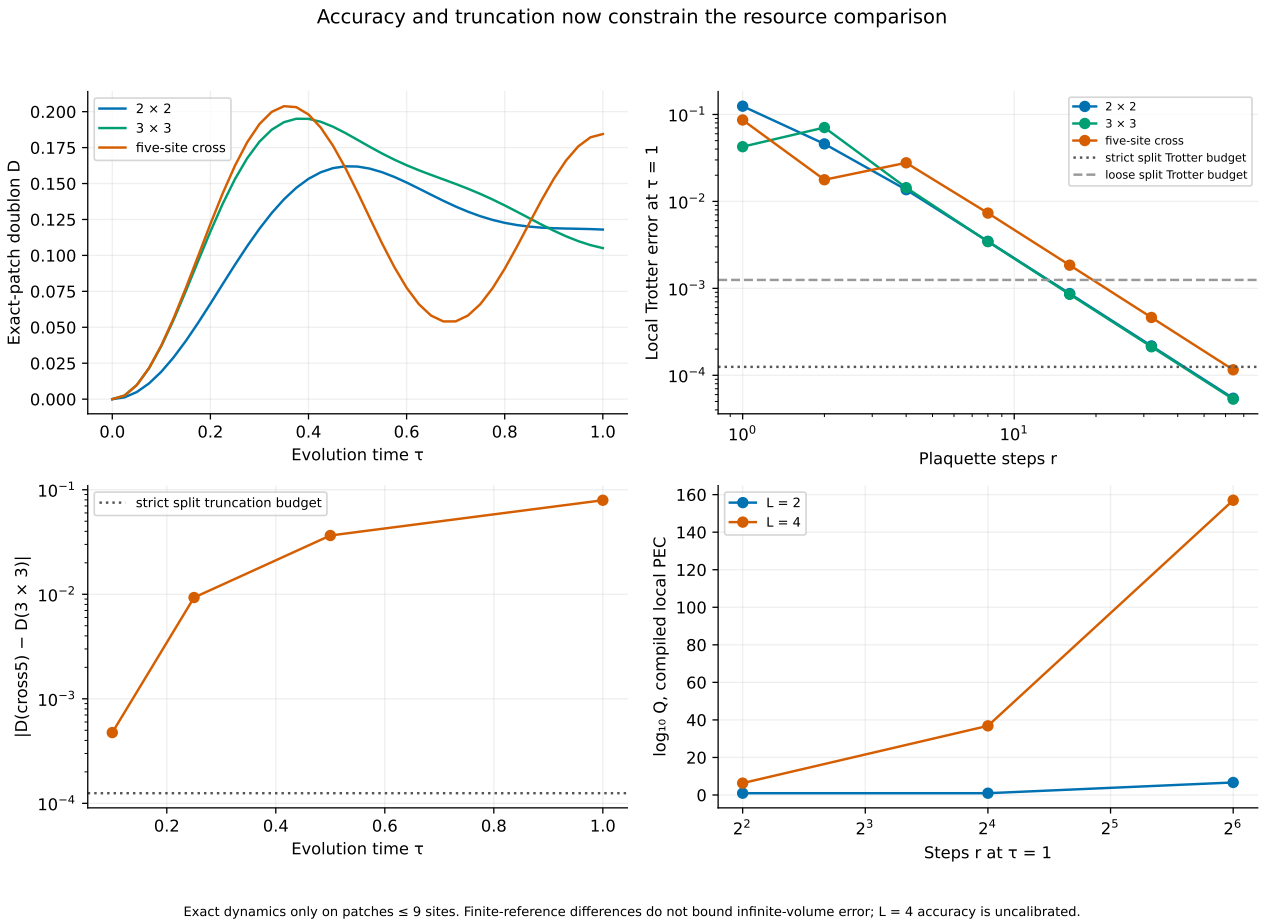


Figure 2: Exact patch dynamics, local product-formula errors, a nested spatial comparison, and the cost sensitivity to r. Each exact curve belongs to its own finite patch. The spatial difference is not an infinite-volume error certificate. The $L = 4$ cost sweep does not assume the small-patch error law holds at $L = 4$.

Primitive idle channels, initial preparation flips, and final readout flips are also included. A direct Pauli-mixture implementation independently checks the optimized channel calculation. For selection event A, the measured quantity is $D_A = \mathbb{E}[D\mathbf{1}_A]/\Pr(A)$; a large or small acceptance alone says nothing sufficient about its bias.

At $r = 4$, the unselected noise bias relative to the ideal Trotter circuit is 0.020371. Selecting the two spin parities reduces it to 0.005808 with acceptance 0.81958. Selecting exact $N_\uparrow = N_\downarrow = 2$ reduces it to 0.004212 with acceptance 0.80943. Its bias relative to exact dynamics, including the independently measured Trotter error, is 0.01785. At $r = 64$, the number-selected noise bias rises to 0.1070. Postselection helps here, but by itself does not meet the strict precision target.

The parity acceptance agrees with the independently propagated Pauli-character formula. Trace errors are below 1.4×10^{-12} and the calculated density matrices remain positive. These results apply to the stated eight-qubit circuits and noise channels. They do not prove failure of selection at $L = 4$ or of combinations with other mitigation methods. The larger trapped-ion experiment [4] remains a separate circuit, geometry, and noise comparison.

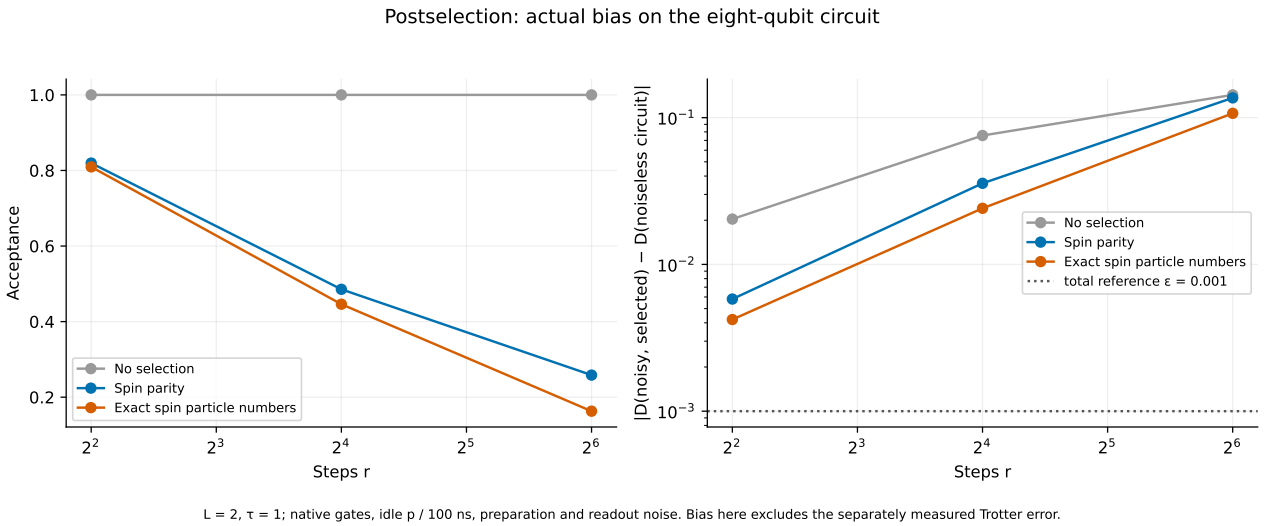


Figure 3: Exact acceptance and actual observable bias under the declared eight-qubit noise model. The right panel isolates noise bias from Trotter bias. No extrapolation from these curves to $L \geq 4$ is used.

5 Cultivation for both FT and hybrid supply

Gidney, Shutty, and Jones report cultivated-state error approximately 2×10^{-9} at $p = 10^{-3}$ and, in their Fig. 15, expected volume 6.6×10^4 qubit-rounds for the distance-five protocol with early rejection and sharing [5]. The output is in a grafted matchable code after escape to nominal distance 15; its behavior is not identical to an ordinary distance-15 surface-code patch. Holding, transfer, growth, and consumption require additional accounting.

Our declared packed-supply scenario rounds the expected volume to $V = 10^5$ qubit-cycles, takes footprint $q_f = 1024$, successful latency $\ell_f = 50$ cycles, and charges a $2d$-cycle handoff. These latter choices are supply assumptions, not a compiled factory. The comparison also uses an unshared fixed-slot volume $V = 1024 \times 50/0.01 = 5.12 \times 10^6$. Mapping the paper's uniform physical noise and round definitions to this device's $p_{1q} = p/10$ and 240 ns cycle remains conditional.

For handoff error q_h, define

$$q_{\text{ready}} = 2 \times 10^{-9} + q_h, \qquad s_{\text{raw}} = \max(V/q_f, \ell_f) + 2d. \tag{9}$$

We sweep $q_h = 0, 10^{-9}, 10^{-8}$. Direct consumption is permitted only when

$$N_T q_{\text{ready}} \Gamma \leq \epsilon_{\text{magic}}, \qquad \epsilon_{\text{magic}} = 0.075\epsilon, \tag{10}$$

with $\Gamma = 1$ for FT and the retained conservative analog-mitigation amplification for hybrid. Otherwise one additional 15-to-1 stage is available, with the same reference coefficients used previously [6]:

$$q_{\text{out}} = 35 q_{\text{ready}}^3, \qquad s_{\text{out}} = \frac{15 s_{\text{raw}} + 11d}{1 - 15 q_{\text{ready}}}. \tag{11}$$

The area includes the source footprint, one handoff buffer per source, and twelve further tiles if this stage is used. Surface-memory faults of the data, buffers, and final distillation stage enter the logical budget; the source preparation error already appears in Eq. (10). Supply and consumption run concurrently, so duration contains their maximum, and source copies and code distance are optimized under the week cutoff.

At strict precision and zero additional handoff error, direct unamplified consumption supports at most 37500 states. The compiled $L = 4$ FT count, 19584, fits; the $L = 6$ circuit requires the extra stage. Taking $q_h = 10^{-8}$ forces that stage already at $L = 4$. Cultivation therefore does not remove a distillation level independently of the task budget. In particular, the roughly 10^5 cutoff discussed around TopQAD Fig. 16 counts small-angle rotations at about 100 T states per rotation, not 10^5 T states [7].

$L = 4$, $r = 4$ reference	Q, $\epsilon = 10^{-3}$	Q, $\epsilon = 10^{-2}$
Serial STAR	166698	45486
FT, conventional serial factory	13791250	216890
FT, packed cultivation, $q_h = 0$	748656	52180
Hybrid, conventional serial factory	669714	54150
Hybrid, packed cultivation, $q_h = 0$	57472	47232

The conventional reference uses injected-state error $46p/15$. All rows remain conditional on $r = 4$. The loose-precision FT rows reuse the stricter synthesized words; resynthesis could improve them further. Cultivation changes the supply comparison appreciably, but neither these numbers nor their ranking should be promoted into a result for accurate dynamics.

6 Explicit STAR retries, covariance, and parallelism

The logical injection instrument prepares $|A_\theta\rangle = e^{-i\theta Z}|+\rangle$, applies $\mathrm{CNOT}_{\mathrm{data}\to\mathrm{ancilla}}$, and measures the ancilla in Z. Its data Kraus operators are

$$K_0 = 2^{-1/2}e^{-i\theta Z}, \qquad K_1 = 2^{-1/2}e^{+i\theta Z}. \tag{12}$$

After outcome one, an injection of angle 2θ corrects a successful next branch: $K_0(2\theta)K_1(\theta) = K_0(\theta)/\sqrt{2}$. Repeating this gives the standard angle-doubling protocol underlying the STAR reference [8, 9]. CUDA-Q P0 now contains explicit preparation, CNOT, measurement, conditional retries, and a returned exhaustion flag for caps $1, 2, 4, 8, 24$. This compilation does not supply a surface-code injection or patch-growth implementation.

For a Pauli fault on either wire immediately before or after CNOT, each conditional K remains diagonal or antidiagonal with equal-magnitude entries. Hence

$$K^\dagger K = I/2, \qquad KZ = \pm ZK. \tag{13}$$

The first identity makes the outcome probability independent of the data state; the second makes the conditional map commute with the channel $\mathcal{Z}(\rho) = Z\rho Z$. It therefore commutes with a Z-dephasing inverse, which is a linear combination of $\mathcal{I}$ and $\mathcal{Z}$. Mixtures of such faults and compositions over retries retain this local covariance under the stated branch-independent Pauli model. All sixteen two-qubit Paulis at both locations are checked independently.

This verifies the needed local commutation condition for that declared instrument. It does not establish branch-independent residual noise for physical state preparation, patch growth, measurement-dependent selection, or a mismatched channel inversion. The worst-case Γq_L bias bound therefore remains the default; q_L is a separately labeled covariance-based reference scenario. The physical STAR schedule still has to justify using it.

For b simultaneous ideal injections with geometric success probability $1/2$ and cap K, the mean batch retry time in units of one attempt is

$$\mathbb{E}[M_b] = \sum_{k=0}^{K-1} \Pr(M_b > k) = \sum_{k=0}^{K-1} \left[1 - (1 - 2^{-k})^b\right]. \tag{14}$$

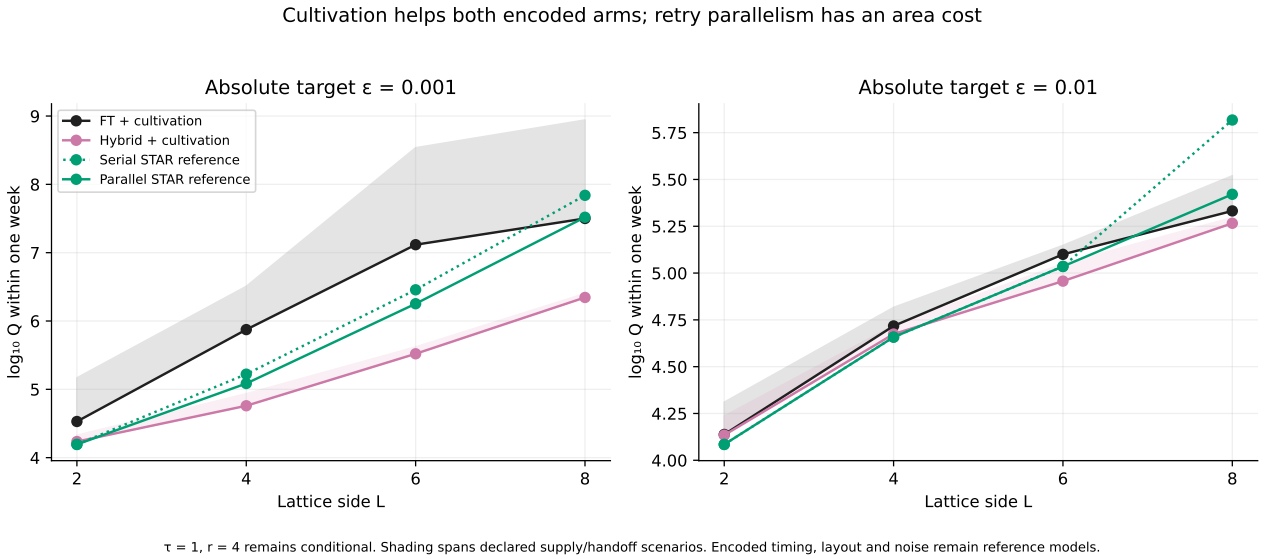


Figure 4: Cultivation is offered to both FT and hybrid, with source-copy and distance optimization. Shading spans declared handoff and supply scenarios, not a statistical confidence interval. Parallel STAR includes retry tails, support conflicts, and lane area. All curves retain the accuracy-unvalidated $r = 4$ workload and reference encoded layouts.

It approaches 2 for one lane, but is approximately 3.50 for four lanes. Simply dividing the serial duration by four misses the slowest retry. The implemented policy has a barrier between batches, reserves six injection tiles per lane, and assumes nine clocks per attempt, recovering the previous 18-clock serial mean. It forms disjoint batches using full JW supports, not merely disjoint fermionic endpoints. An asynchronous policy could improve this conservative batch schedule.

With R rotations the ideal no-exhaustion probability is $(1 - 2^{-K})^R$. We choose K from a small allocated tail allowance and charge the acceptance penalty. At $L = 4$ and strict precision, two lanes minimize area at $Q = 121716$, versus 166698 for the earlier serial reference. Four and eight lanes reduce attempt time further but lose on area. At loose precision one lane still wins. These results constrain the particular logical mapping and resource model, not the best possible STAR architecture.

7 Compiler diagnostics and what remains unresolved

The rejected FT backend timing is now exercised by a smaller circuit: four fixed-angle Pauli rotations repeated four times, totaling 16 T states. The installed CUDA-Q Logical scheduler rejects it because a nested entry extends outside its parent interval. The 4-, 8-, and 32-T versions pass a loose timing bound. Separate two-resource template-call controls all return their expected durations. Exact inputs, P3 IR, and the failure diagnostic are retained. This reduces the failure without identifying its root cause; the passing cases are not a calibration of encoded timing.

The 73 tests cover fermionic permutation signs, stage identity, observable pruning, independent exact and product-formula dynamics, explicit Pauli-noise channels, logical STAR algebra, and magic-state budget switching. Saved native circuits carry source hashes; the manifest archives sources and hashes all new artifacts. These checks establish the specified calculations, not hardware performance.

The immediate scientific dependencies are now explicit. A larger-patch or controlled local calculation must qualify spatial truncation and r; the same qualified circuit must then be compiled for every architecture. Encoded refinement needs physical STAR injection/growth and cultivation

handoff schedules, with residual-channel validation. Qubitization remains a separately identified later algorithm comparison.

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